

Pitch to Postgres: Designing and Querying a Relational Database for FIFA World Cup Data

DS604 Introduction to Data Management
Milestone 1: Dataset and Tentative Queries

Group member	Student ID
Devanshi Dudhatra	202618027
Pujita Sunapu	202618003
Rahul Saha	202618037

Submitted 3 October 2026

Data: The Fjelstul World Cup Database by Joshua C. Fjelstul (CC-BY-SA 4.0)

1. Project overview

Aim. Build a relational database for the complete history of the men's FIFA World Cup in PostgreSQL, load a real public dataset into it, and use SQL to answer analytical questions about teams, players, goals, discipline, venues, managers and long-term trends.

Why this dataset. World Cup data is familiar enough that results can be checked against well-known facts, which lets us validate our SQL. It is also rich enough to need a real relational design: 27 linked tables, composite keys, bridge tables, historical teams that no longer exist, and several data-quality issues to handle.

What we will deliver

Milestone	Due	Deliverable
1. Dataset and tentative queries	03-Oct-2026	This document
2. Schema	15-Oct-2026	Normalised schema, <code>sql/ddl.sql</code> , ER and relational schema diagrams
3. Load notebook	17-Oct-2026	<code>notebooks/03_load_data.ipynb</code> : loads all 27 CSV files with <code>pandas.to_sql</code>
4. Analytical queries	06-Nov-2026	<code>sql/queries.sql</code> and <code>notebooks/04_queries.ipynb</code> with results and interpretation
5. Views	06-Nov-2026	<code>sql/views.sql</code> , with at least five views
Final package	15-Nov-2026	Final report, README, viva notes

Tools. PostgreSQL (the course server, with a local PostgreSQL 18 installation for testing), Python with pandas and SQLAlchemy for loading and querying, and Jupyter notebooks. All SQL is standard and PostgreSQL-compatible. Credentials are read from a `.env` file and never written in code.

Approach. Before choosing queries we profiled every file (`src/profile_data.py`, report in `docs/dataset_understanding.md`). We checked each candidate key and relationship against the data and recorded every inconsistency. The query list below reflects what the data can actually support.

2. Dataset description

Item	Details
Name	The Fjelstul World Cup Database
Author	Joshua C. Fjelstul, Ph.D.
Source	https://github.com/jfjelstul/worldcup
Licence	Creative Commons Attribution-ShareAlike 4.0 (CC-BY-SA 4.0)
Format	27 CSV files (UTF-8), 57,269 data rows in total
File date	July 2022
Actual coverage	21 men's World Cup final tournaments, 1930 to 2018 (13 July 1930 to 15 July 2018). The 2022 tournament is not included. Women's tournaments and qualifying matches are not included.

Item	Details
Content	900 matches, 2,548 goals (52 own goals), 84 national teams (including historical teams), 7,907 players, 380 referees, 357 managers, 185 stadiums in 17 countries
Event detail	Goals, managers and referees for every tournament. Line-ups, substitutions and cards from 1970 (cards were introduced that year). Penalty shoot-outs from 1982 (the first shoot-out).
Our modifications	No data values have been changed. The files were placed unchanged in <code>data/</code> . Cleaning applied during loading (for example turning the text "not available" into SQL NULL) will be listed in Milestone 3.

Attribution. This project uses *The Fjelstul World Cup Database* by Joshua C. Fjelstul (<https://github.com/jfjelstul/worldcup>), licensed under CC-BY-SA 4.0. Our derived outputs are shared under the same licence.

3. The 27 tables, grouped by theme

The *grain* is what one row represents. Each key was checked for uniqueness on the real data, and every key below is unique with no missing values.

3.1 Reference tables

Table	Rows	Grain (one row is...)	Candidate key
confederations	6	a continental confederation (UEFA, CONMEBOL, ...)	confederation_id
teams	84	a national team, including historical teams such as West Germany and the Soviet Union	team_id
stadiums	185	a stadium	stadium_id
awards	8	an award type (Golden Ball, Golden Boot, ...)	award_id

3.2 People

Table	Rows	Grain	Candidate key
players	7,907	a player	player_id
managers	357	a manager (head coach)	manager_id
referees	380	a referee	referee_id

3.3 Tournament structure

Table	Rows	Grain	Candidate key
tournaments	21	a World Cup edition	tournament_id
tournament_stages	107	a stage of a tournament (group stage, round of 16, final, ...)	(tournament_id, stage_number)
groups	117	a group within a group stage	(tournament_id, stage_number, group_name)
host_countries	22	a host nation of a tournament (2002 had two)	(tournament_id, team_id)

3.4 Participation (who took part)

Table	Rows	Grain	Candidate key
qualified_teams	457	a team at a tournament	(tournament_id, team_id)
squads	10,142	a player in a team's squad at a tournament	(tournament_id, team_id, player_id)
manager_appointments	469	a manager leading a team at a tournament	(tournament_id, team_id, manager_id)
referee_appointments	511	a referee appointed to a tournament	(tournament_id, referee_id)

3.5 Matches

Table	Rows	Grain	Candidate key
matches	900	a match (replays are separate matches)	match_id
team_appearances	1,800	one team's side of a match (two rows per match)	(match_id, team_id)

3.6 Match events

Table	Rows	Grain	Candidate key
goals	2,548	a goal	goal_id
bookings	2,466	a card shown to a player (1970+)	booking_id
substitutions	6,464	one player going off or coming on (two rows per substitution, 1970+)	substitution_id
penalty_kicks	279	a kick in a penalty shoot-out (1982+)	penalty_kick_id
player_appearances	18,623	a player who played in a match (1970+)	(match_id, player_id)
manager_appearances	1,842	a manager in charge of a team in a match	(match_id, team_id, manager_id)
referee_appearances	900	the referee of a match (one per match)	match_id

3.7 Results and standings

Table	Rows	Grain	Candidate key
group_standings	458	a team's final line in a group table	(tournament_id, stage_number, group_name, position)
tournament_standings	84	a top-four finishing position in a tournament	(tournament_id, position)

3.8 Awards

Table	Rows	Grain	Candidate key
award_winners	132	a player winning an award at a tournament (ties give several rows)	(tournament_id, award_id, player_id)

4. Expected relationships

Every relationship below was checked against the data: all of them have **zero orphan values**, meaning every referenced id exists in the parent table.

One-to-many (1:N)

- A confederation has many teams and many referees.
- A tournament has many stages, and a group stage has many groups. Each group has one table in `group_standings`.
- A tournament stage has many matches, and a stadium hosts many matches.
- A match has many goals, bookings, substitutions and (if decided on penalties) shoot-out kicks.
- A match has exactly one referee (`referee_appearances` is effectively 1:1 with `matches`).

Many-to-many (M:N), resolved by bridge tables

Relationship	Bridge table
tournaments and teams (participation)	<code>qualified_teams</code>
tournaments and teams (hosting)	<code>host_countries</code>
tournaments and teams (top-four finish)	<code>tournament_standings</code>
(tournament, team) and players	<code>squads</code>
matches and teams	<code>team_appearances</code> (exactly 2 per match)
matches and players	<code>player_appearances</code>
(tournament, team) and managers	<code>manager_appointments</code>
matches, teams and managers	<code>manager_appearances</code>
tournaments and referees	<code>referee_appointments</code>
tournaments, awards and players	<code>award_winners</code>

Composite (multi-column) references. Event tables can reference `squads` on (tournament_id, team_id, player_id) rather than only `players`. This is a stronger rule: a goal scorer must be in that team's squad for that tournament, and the data satisfies it for every goal, card, substitution, shoot-out kick, appearance and award.

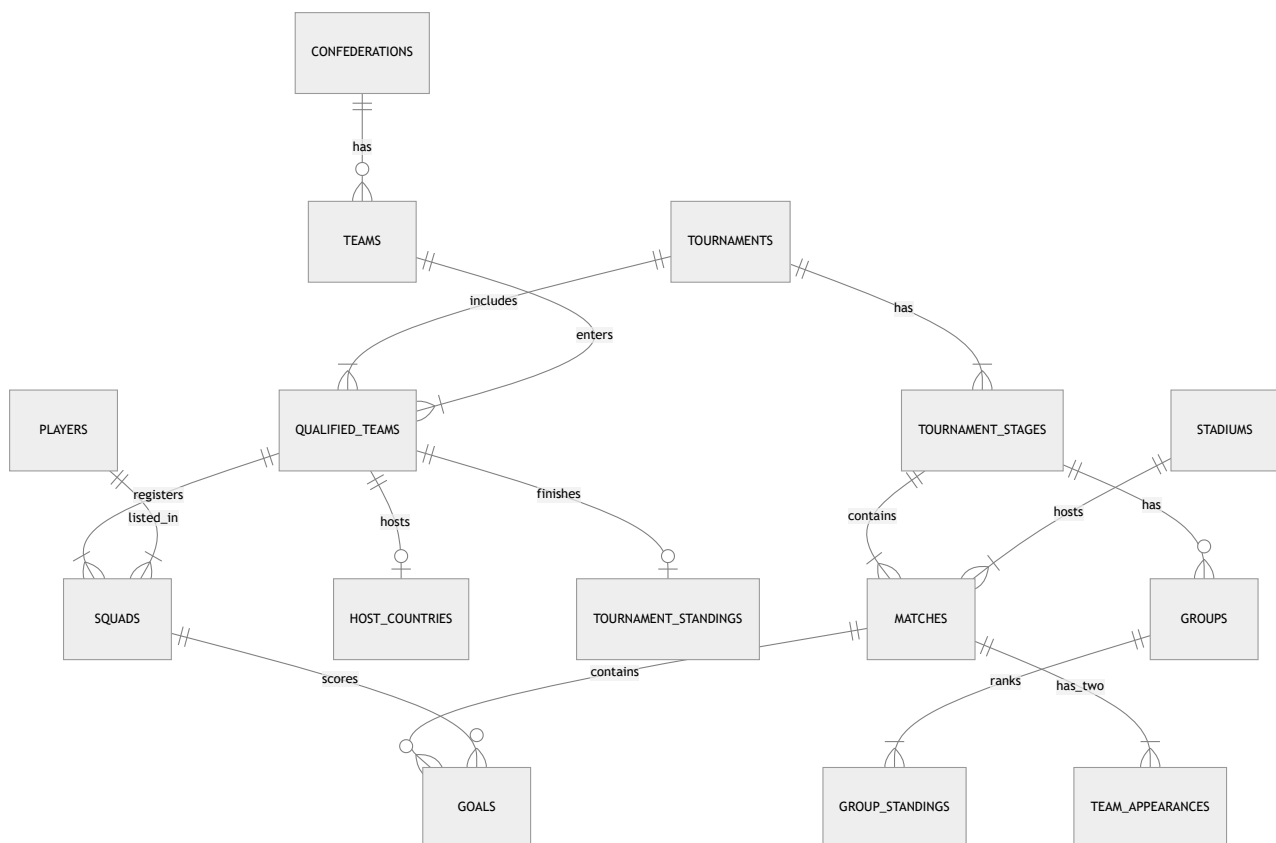


Figure 1: Core relationships (simplified; the full ER and relational schema diagrams come in Milestone 2).

Points that affect the design (found during profiling)

1. `matches` does not store `stage_number`, so it links to `tournament_stages` through (`tournament_id`, `stage_name`). `groups` and `group_standings` use different stage names for 1950 and 1974 to 1982 (for example "first group stage" where `tournament_stages` says "group stage"). Joins must therefore use `stage_number`, not the name.
2. Germany and West Germany are separate teams with the same team code (DEU). Team code is therefore not a key.
3. On 12 occasions a team had joint managers at a tournament (42 team-matches), so the manager tables need `manager_id` in their keys.
4. Many tables repeat names copied from their parent tables (`team_name`, `tournament_name`, ...). These copies agree with the parent tables everywhere, so the normalised schema can drop them.

5. Tentative analytical queries

Thirty queries, grouped by theme. "Concepts" lists the main SQL techniques each query will demonstrate. Unless stated otherwise, wins, draws and losses are worked out from the score. A match settled on penalties counts as a draw, which is how FIFA records it; the dataset's own `result` column records it as a win.

5.1 Team performance

#	Business question	Concepts	Main tables
Q1	Which teams are the most successful, counting titles, final appearances (top two) and top-four finishes?	JOIN, GROUP BY, SUM(CASE ...), ORDER BY	tournament_standings, teams
Q2	What is each team's all-time record (played, won, drawn, lost, goals, win %), for teams with at least 10 matches?	GROUP BY, HAVING, CASE, ratio arithmetic	team_appearances, teams
Q3	Do host nations do better than everyone else? Compare their win rate and how far they go.	CTE, LEFT JOIN, CASE, UNION ALL	host_countries, team_appearances, tournament_standings
Q4	How does each confederation perform, decade by decade (win rate and top-four finishes)?	Multi-table JOIN, GROUP BY on an expression (decade), CASE	team_appearances, teams, confederations, tournaments
Q5	Which teams have never got past the opening round of any tournament they played in?	CTE, MAX(stage_number), GROUP BY / HAVING, NOT EXISTS	team_appearances, tournament_stages, teams
Q6	Which teams have finished in the top four but never won the title?	Set operation (EXCEPT)	tournament_standings, teams
Q7	What is the head-to-head record between two chosen teams (for example Brazil and Sweden)?	Self-join on team_appearances, parameters, aggregation	team_appearances
Q8	How did each country's title count build up over time?	Window function: running total, SUM() OVER (PARTITION BY ... ORDER BY ...)	tournament_standings, tournaments
Q9	Which teams most often came back to win after conceding the first goal?	CTE, ROW_NUMBER() to find each match's first goal, JOIN	goals, team_appearances

5.2 Player performance

#	Business question	Concepts	Main tables
Q10	Who are the all-time top 10 goal scorers (excluding own goals)?	JOIN, WHERE, GROUP BY, ORDER BY / LIMIT, RANK()	goals, players
Q11	Who was the top scorer at each tournament, and does that match the official Golden Boot winner?	CTE, RANK() OVER (PARTITION BY tournament), LEFT JOIN	goals, players, award_winners
Q12	Which players scored hat-tricks, and who scored the most goals in one match?	GROUP BY, HAVING COUNT(*) >= 3, JOIN	goals, players, matches
Q13	How many goals were scored by substitutes in each tournament since 1970, and what share is that?	JOIN on (match_id, player_id), CASE, GROUP BY	goals, player_appearances
Q14	What is the average player age by position and tournament?	Date arithmetic (AGE, EXTRACT), AVG, NULL handling	squads, players, tournaments
Q15	Who has made the most World Cup appearances for each nation (1970+)?	ROW_NUMBER() OVER (PARTITION BY team), CTE	player_appearances, players, teams
Q16	Which players have played for more than one national team, and which teams?	GROUP BY, HAVING COUNT(DISTINCT ...), STRING_AGG	squads, players, teams

5.3 Goals and scoring patterns

#	Business question	Concepts	Main tables
Q17	How many goals per match were scored at each tournament, and how did that change from the previous edition?	GROUP BY, window function LAG()	matches, tournaments
Q18	What are the biggest winning margins and the highest-scoring matches?	Derived columns, DENSE_RANK(), ORDER BY	matches
Q19	When are goals scored? Count goals by 15-minute period, plus extra time and stoppage time.	CASE bucketing, GROUP BY	goals
Q20	How have the shares of penalty goals and own goals changed over time, especially in 2018 (the first World Cup with VAR)?	Conditional aggregation (SUM(CASE ...)), GROUP BY	goals, tournaments
Q21	What are the penalty shoot-out records: shoot-outs won and lost per team, and kick conversion rates?	CTE, JOIN, GROUP BY, HAVING	penalty_kicks, team_appearances

5.4 Discipline and refereeing (1970 onwards)

#	Business question	Concepts	Main tables
Q22	Which referees are the strictest (cards per match, minimum five matches)?	JOIN, GROUP BY, HAVING, subquery for the overall average	bookings, referee_appearances, referees
Q23	How have cards per match changed over time? Show a three-tournament moving average.	Window function AVG() OVER (ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)	bookings, matches
Q24	Are knockout matches more ill-tempered than group matches (cards and sendings-off per match by stage)?	CASE, GROUP BY, LEFT JOIN so that matches with no cards still count	bookings, matches
Q25	Which teams have received the most cards and sendings-off?	GROUP BY, HAVING, ORDER BY	bookings, teams

5.5 Venues and hosting

#	Business question	Concepts	Main tables
Q26	Which stadiums and cities have hosted the most World Cup matches?	JOIN, COUNT, DENSE_RANK()	matches, stadiums

5.6 Managers

#	Business question	Concepts	Main tables
Q27	Which managers have managed the most matches and won the most?	Three-table JOIN on a composite key (match_id, team_id), GROUP BY	manager_appearances, team_appearances, managers
Q28	Do teams with a foreign manager go further than teams with a manager of their own nationality?	CASE, correlated subquery or CTE, AVG	manager_appointments, teams, team_appearances, tournament_stages

5.7 Awards

#	Business question	Concepts	Main tables
Q29	How often does the Golden Ball (best player) go to a player from the champion team?	LEFT JOIN, CASE, IS NULL test	award_winners, tournament_standings

5.8 Data quality and integrity

#	Business question	Concepts	Main tables
Q30	Does every goal in the goals table match the scores in the matches table? List any team-match where they differ.	CTE, LEFT JOIN, COALESCE, set comparison (EXCEPT)	goals, team_appearances

5.9 SQL concepts covered

Concept	Queries
Inner joins (2 or more tables)	Q1, Q2, Q4, Q10, Q12, Q13, Q14, Q22, Q26, Q27
Outer joins (LEFT JOIN)	Q3, Q11, Q24, Q29, Q30
Self-join	Q7
GROUP BY with HAVING	Q2, Q5, Q12, Q16, Q21, Q22, Q25
Subqueries (scalar, correlated, NOT EXISTS)	Q5, Q22, Q28
Common table expressions (WITH)	Q3, Q5, Q9, Q11, Q15, Q21, Q28, Q30
Window: RANK / DENSE_RANK	Q10, Q11, Q18, Q26
Window: ROW_NUMBER	Q9, Q15
Window: LAG	Q17
Window: running total	Q8
Window: moving average	Q23
Set operations (EXCEPT, UNION ALL)	Q3, Q6, Q30
CASE logic and conditional aggregation	Q1, Q2, Q3, Q4, Q13, Q19, Q20, Q24, Q28, Q29
Date functions	Q14
String aggregation	Q16

6. Changes to the starting list, and why

We kept all 16 starting queries. They map to our numbering as follows:

Starting list	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Our query	Q1	Q17	Q3	Q4	Q5	Q10	Q11	Q20 + Q21	Q13	Q14	Q22	Q23	Q26	Q7	Q8	Q18

After profiling the data we made these adjustments:

Starting query	Change	Reason (from the data)
1. Most successful teams	Finals are counted as finishing positions 1-2 in <code>tournament_standings</code> , not as matches labelled "final".	The 1950 World Cup had no final match. It ended with a four-team final round, so a "final" label would miss Uruguay and Brazil in 1950.
5. Never left the group stage	Redefined as "never got past the opening round", based on the stage order (<code>stage_number</code>).	The 1934 and 1938 tournaments had no group stage; teams went out in an opening round of 16. Using the "group stage" label leaves out Egypt and the Dutch East Indies (30 teams with the new rule against 28).

Starting query	Change	Reason (from the data)
8. Penalty conversion and shoot-out records	Split into Q20 (share of goals from in-match penalties) and Q21 (shoot-out records and kick conversion).	Missed in-match penalties are not in the data, only scored ones, so in-match conversion cannot be calculated. Shoot-out kicks (1982+) include misses, so shoot-out conversion can.
9. Goals by substitutes	Limited to 1970 onwards.	Line-ups and substitutions exist only from 1970.
11, 12. Referees and bookings	Limited to 1970 onwards.	Cards were introduced in 1970, and the bookings table starts then.
3, 4, 14 (and Q2, Q27)	Wins and draws are taken from the score, not the <code>result</code> column.	In all 30 shoot-out matches the score is level, but <code>result</code> records a home or away win.
14. Head-to-head	Implemented as a self-join of <code>team_appearances</code> .	Each match appears from both sides, so pairing a team's row with its opponent's row is a natural self-join.
4. Confederation by decade	Kept, with a stated caveat.	The confederation in <code>teams</code> is the current one. Australia is listed in the AFC, although it qualified as an Oceania (OFC) member in 1974 and 2006.

Queries we added (Q2, Q6, Q9, Q12, Q15, Q16, Q19, Q24, Q25, Q27, Q28, Q29, Q30) come from the insights framework in `docs/dataset_understanding.md`. They fill gaps in themes and SQL concepts: an overall team record (Q2), set operations (Q6, Q30), ROW_NUMBER (Q9, Q15), managers (Q27, Q28), awards (Q29) and an integrity check (Q30) that tests the loaded database.

Ideas we considered and dropped, because the data cannot support them: in-match penalty conversion (missed penalties not recorded), assists (not recorded), attendance (only stadium capacity is recorded), fastest goals in seconds (minutes only), and possession or shots (not recorded).

7. Data caveats the queries must respect

- **Coverage:** 1930-2018 only. Line-ups, cards and substitutions from 1970; shoot-outs from 1982.
- **Missing values:** the text values "not available" and "not applicable", and shirt number 0, mean missing. They will become SQL NULL during loading. Birth dates are missing for 78 players, who are left out of age calculations.
- **Historical teams:** West Germany and Germany, the Soviet Union and Russia, Yugoslavia, Serbia and Montenegro and Serbia, and Czechoslovakia and the Czech Republic are separate teams. Where we combine them we will say so; for example, Germany's four titles include West Germany's three.
- **Special cases:** four drawn matches were replayed (1934, 1938), one 1938 tie was a walkover, there were group play-off matches in 1954 and 1958, and there was no third-place match in 1930 or 1950. There are no shared third places and no abandoned matches.
- **Points systems:** 2 points per win up to 1990 and 3 points from 1994, so raw points cannot be compared across eras.

Full details of all of these are in `docs/dataset_understanding.md`.

8. Next steps

- **Milestone 2:** turn the verified keys into a normalised PostgreSQL schema with primary and foreign keys and constraints, then draw the ER and schema diagrams.
- **Milestone 3:** load all 27 files with `pandas.to_sql(if_exists="append")` into the tables created by the DDL, then check the row counts.
- **Milestone 4:** write the 30 queries and check at least 10 results against known World Cup facts.

Data: Joshua C. Fjelstul, The Fjelstul World Cup Database, <https://github.com/jfjelstul/worldcup>, CC-BY-SA 4.0. No data values were modified. This document is a derived work under the same licence.